Michael Remington Coulter

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PROFESSIONAL EXPERIENCE

Arm | Austin, TX

Senior Software Engineer

Apr 2023 - Present

Develop Interconnect Architect, Arm's Python/TypeScript SoC interconnect configuration tool for internal engineers and external customers, alongside AI tools for engineering workflows.

- Implemented Python Model Context Protocol (MCP) tools compiling typed intent into dependency-ordered execution plans, with checkpoints, semantic checks and rollback.
- Validated creation, import, modification, diagnosis and rollback in two bounded agent evaluations; rollback restored the exact configuration hash. 92 live acceptance checks passed.
- CMN GUI Lead: created and own all coherent-mesh GUI tooling from CMN-800 onward, including device/group configuration, custom routing and deadlock investigation.
- Built reproducible verification for packaged MCP tools in clean macOS/Windows Codex sessions, checking payload integrity, provider routing, handshake and cleanup without inherited context.
- Automated Jenkins/LSF failure replay at the original revision, patch transfer and repair validation; gated flaky-test fixes on three consecutive clean replays.
- Developed a Rust/Tauri 2 workbench for repository, build and process control, with a React/TypeScript UI, shared desktop/CLI core and bounded asynchronous Git inspection.
- Created reproducible documentation generation and task-to-code indexes to give agents source-aware context for navigating an Angular/Python application.
- Integrated Python configuration policies with Angular editors, coordinating versioned state, checkpoints, recovery and editing permissions; invalidated compliance after design changes.

Recognition: Most Innovative Award, Engineering AI Showcase (Dec 2025): multi-agent orchestration and context-gathering system.

Software Engineer

Apr 2020 - Apr 2023

- Built Flask services for Jupyter notebook lifecycle and AWS EFS persistence in Pelion's embedded-ML platform; designed PostgreSQL schemas and end-to-end tests.

Graduate Software Engineer

Jan 2019 - Apr 2020

- Built Verilog FPGA modules and Mbed OS firmware to test board I/O; contributed to test-shield PCBs. Arm Quality Award (team, 2020).

TECHNICAL SKILLS

Languages: Python, Rust, TypeScript/JavaScript, SQL, Verilog
Agent / validation: MCP, typed plans, semantic checks, rollback, Codex, session isolation
Developer systems: Git, Jenkins, Playwright, LSF, RHEL 8; React, Tauri 2, Flask, PostgreSQL

EDUCATION

The University of Texas at Austin
Bachelor of Science, Electrical and Computer Engineering | Dec 2018
Primary focus: Computer Architecture and Embedded Systems | Secondary: Integrated Circuits